

3D DATA PROCESSING - LAB 4

Point Cloud Semantic Segmentation using Sparse Convolutions (MinkUNet)- Report

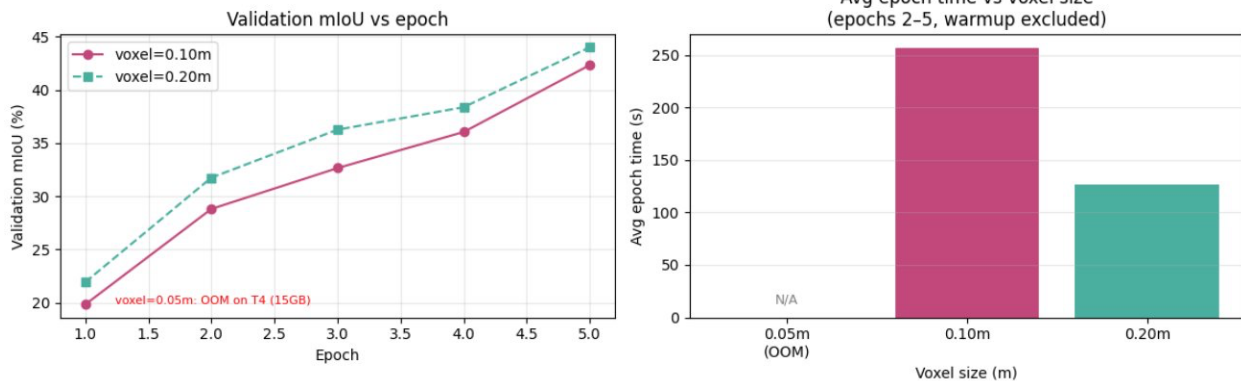
Task 4.1: Voxel Size Sensitivity Study

All runs use `use_remission=True` and `augment=True` for 5 epochs with `batch_size=8`. The average epoch time is computed over epochs 2–5. Epoch 1 is omitted as it includes DataLoader worker startup overhead (~600s vs ~255s after it stabilizes).

Summary Table

Voxel size [m]	Average epoch time [s]	Final validation mIoU [%]	Test mIoU [%]
0.05	OOM (T4, 15 GB)	---	---
0.10	256.98	42.33	26.90
0.20	126.91	44.01	19.30

Plots: Validation mIoU and epoch time



Analysis

- **voxel_size = 0.05 m**

At 5 cm resolution the spatial grid is $2000 \times 2000 \times 200$ voxels. A typical SemanticPOSS scan produces approximately 130.000 active voxels, which scales to roughly one million voxels per batch of 8. The index-pair table allocated by *spconv* for a single *SubMConv3d* layer (shape [2, 27, M]) requires ~220 MB per forward pass at this voxel count, **exhausting the 15 GB of VRAM** available on a T4 GPU before completing even one training iteration. Training at this resolution **requires a GPU with at least 24 GB VRAM or a reduced batch size**.

- **voxel_size = 0.10 m**
At 10 cm resolution the grid is $1000 \times 1000 \times 100$ voxels. This is the **best-performing configuration** on the held-out test set (26.90% test mIoU), indicating that this resolution provides sufficient geometric detail for the network to learn discriminative features that generalize to unseen data. Average epoch time is ~257 s.
- **voxel_size = 0.20 m**
At 20 cm resolution the grid is $500 \times 500 \times 56$ voxels, roughly 8x fewer voxels than 0.10 m, since volume scales cubically. This makes training approximately 2x faster (~127 s/epoch). However, coarse voxelization merges nearby points that belong to different semantic classes into a single voxel, **losing fine-grained geometric detail**. Despite reaching the highest validation mIoU (44.01%), the test mIoU drops sharply to 19.30% (a 7.6 pp gap relative to 0.10 m). This gap indicates **overfitting**: at coarse resolution the network sees fewer distinct spatial patterns per epoch and memorizes training-set statistics rather than learning generalizable features. Small objects are particularly affected because their geometry collapses to just a handful of voxels at 20 cm.

In summary, voxel size controls the *accuracy-efficiency* trade-off. Too fine exhausts memory and computation; too coarse loses spatial detail and hurts generalization. For this dataset and hardware, 0.10 m represents the optimal balance.

Task 4.2: Feature Importance (Remission vs. Geometry Only)

Both models are trained for 5 epochs with `voxel_size=0.10` and `augment=True`.

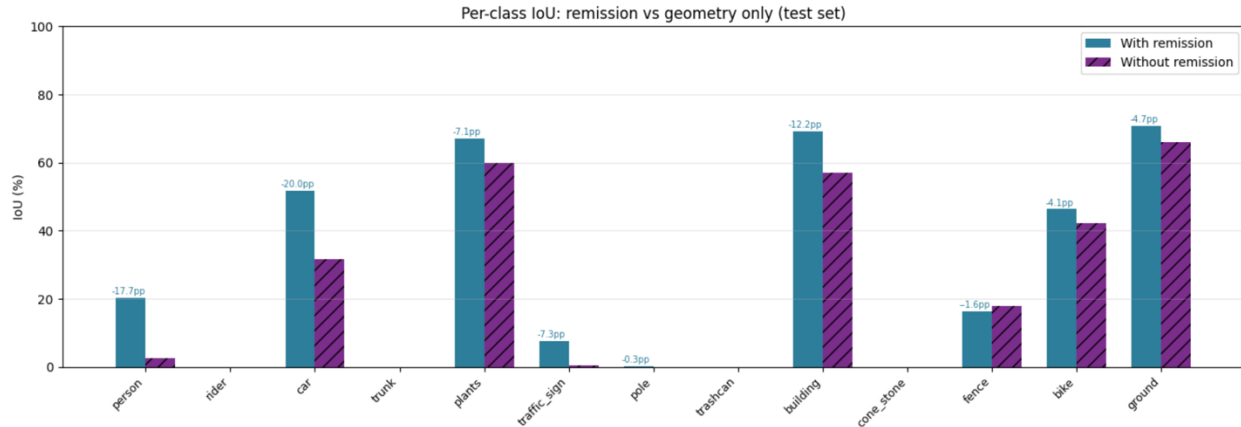
- **Model A** uses the remission (intensity) channel as an additional input feature.
- **Model B** replaces it with a constant value of 1 for all points, making the input geometry-only.

Class-Wise Test IoU comparison

Class	Model A [%]	Model B [%]	Delta [pp]
person	20.26	2.52	-17.74
rider	0.00	0.00	0.00
car	51.67	31.65	-20.02
trunk	0.00	0.00	0.00
plants	67.03	59.89	-7.14
traffic_sign	7.69	0.36	-7.33
pole	0.33	0.00	-0.33
trashcan	0.00	0.00	0.00
building	69.28	57.10	-12.18
cone_stone	0.00	0.00	0.00

fence	16.34	17.89	+1.55
bike	46.33	42.26	-4.07
ground	70.73	66.02	-4.71
mIoU	26.90	21.36	-5.54

Plot



Class-wise analysis and Physical Explanation

The classes with the largest performance drop when remission is removed are:

- car: -20.0 pp (51.67% → 31.65%)**
 Cars have consistent surface reflectance regardless of color or lighting. This makes remission a reliable cue for distinguishing cars from surrounding objects. Without it, the network must segment cars purely from their shape which is easily confused with other large objects.
- person: -17.7 pp (20.26% → 2.52%)**
 People produce inconsistent intensity returns due to the variety of clothing materials and colors. However, the contrast between a person's body and the surrounding ground plane creates a localized intensity pattern that the network can exploit. Without remission, the network must detect people purely from their vertical cylindrical geometry, which at 10 cm voxel resolution captures only a coarse approximation of the human shape.
- building: -12.2 pp (69.28% → 57.10%)**
 Buildings have relatively uniform reflectance. The drop is significant but smaller than car or person because building geometry (large surfaces with vertical orientation) is already highly discriminative.
- traffic_sign: -7.33 pp (7.69% → 0.36%)**
 Traffic signs are retro-reflective by design, producing extremely high intensity returns compared to all other objects in the scene. This makes remission the most informative feature for this class. Without it, the network loses almost all ability to detect signs, as their small geometric footprint is insufficient.

- **plants: -7.1 pp (67.03% → 59.89%)**

Plants show a drop of 7.14 pp without remission, similarly to traffic signs. However, plants were already being segmented reasonably well; the class is spatially extensive and produces dense voxel clusters that give the network sufficient signal even without intensity information.

- **ground: -4.71 pp (70.73% → 66.02%)**

Ground is the highest-scoring class in both configurations due to its spatial extent and horizontal orientation. Without remission, the network loses the ability to distinguish asphalt from painted road markings: asphalt returns low-intensity pulses while retro-reflective lane markings return significantly higher intensity. These two surface types are geometrically identical flat horizontal planes, making remission the only feature that distinguishes them. Despite this, the overall ground IoU remains high because most ground voxels are unmarked asphalt, which the network segments correctly on geometry alone.

The **fence** class shows a **slight improvement without remission** (+1.55 pp). This could be explained because fence geometry (regularly spaced structure) is intrinsically distinctive, and remission provides little information beyond what geometry already captures. However, this is only a hypothesis; the difference is small enough to be attributed to random variation, and the result may reverse with longer training or a larger dataset.

Classes that are 0% in both runs (**rider, trunk, trashcan, cone_stone**) are underrepresented in the training set.

Conclusion

Remission is most valuable for classes where geometry is ambiguous but surface material is distinctive (traffic signs, cars, people), and least valuable where geometry is already highly discriminative (buildings, bikes). This suggests that the choice of input features should be guided by the geometric ambiguity of the target classes: for datasets with a higher proportion of geometrically similar objects, intensity information becomes increasingly critical.